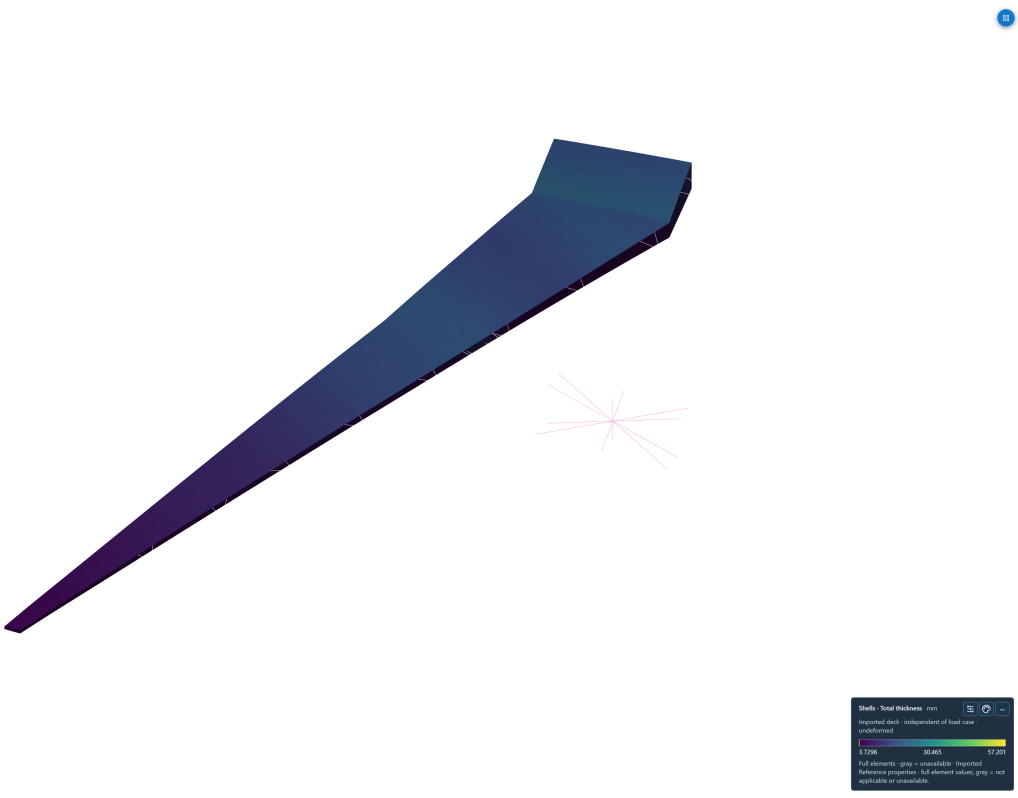
uCRM9 original wingbox

Static cruise loading and the first 20 natural frequencies. The released anisotropic panel properties, engine mass cage, control-surface masses, fuel and constraints are retained.

Model definition	Original coarse structural model
Mesh	23,738 wingbox GRIDs + 74 mass GRIDs; 25,055 CQUADR shells; 242 PSHELL regions and 968 MAT2 material cards.
Sections	Smeared blade-stiffened aluminum panels. Separate membrane, bending, transverse-shear and membrane/bending coupling data are retained. Density 2780 kg/m3.
Concentrated masses	Engine 7501 kg in a 13-point rigid cage; leading/trailing-edge masses 3669 kg; 20% fuel 11000.002 kg. 12 RBE2 and 62 RBE3 connections.
Constraints	Root: 762 GRIDs constrained in 123456. Side of body: 627 distinct GRIDs in DOF 3, plus 11 in DOF 1. Original SPC set 1; 1389 unique constrained nodes.
Load and modal controls	SOL101: fixed nominal-cruise aerodynamic FORCE set plus gravity -9.81 m/s2. SOL103: 20 modes. Original COUPMASS=-1, AUTOSPC=NO, K6ROT=1e5 and WTMASS=1.
JFEM formulation	Native TACS-formulation shell backend with the common rigid-constraint, applied-load and concentrated-mass paths; full singular mass pencil retained.



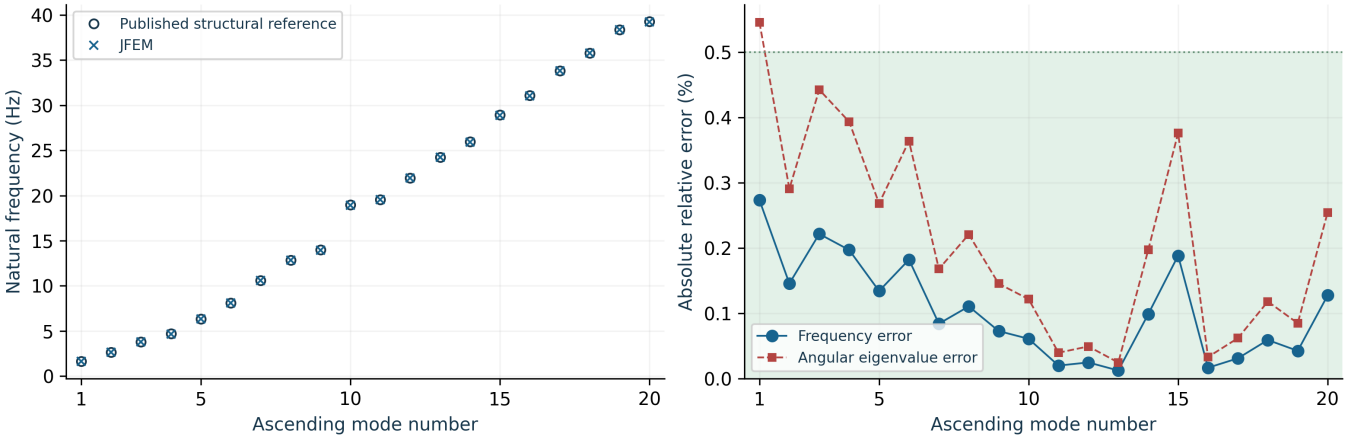
Actual JFEM-Boxer screenshot of the original source model, white background. Color is the original PSHELL thickness in mm. Stiffener effects are smeared; these are not discrete beam sections.

Model and reference data: Brooks, Kenway and Martins, uCRM version 1 (2019), DOI 10.17632/gpk4zn73xn.1, CC BY 4.0. Associated article: AIAA Journal 56(7), 2840-2855 (2018), DOI 10.2514/1.J056603.

Ordered natural frequency comparison

20/20 requested finite modes returned. Maximum error: 0.27339% in frequency, 0.54603% in angular eigenvalue. Within 0.5%: 20/20 frequencies and 19/20 angular eigenvalues.

Mode	Published Hz	JFEM Hz	Frequency error %	Eigenvalue error %
1	1.629275	1.624821	0.2733873	0.5460272
2	2.631756	2.627927	0.1455091	0.2908065
3	3.816032	3.807578	0.2215341	0.4425775
4	4.683773	4.674549	0.1969373	0.3934868
5	6.315011	6.323474	0.1340189	0.2682175
6	8.112136	8.097364	0.182098	0.3638645
7	10.59417	10.58528	0.08395419	0.1678379
8	12.84292	12.82874	0.1104441	0.2207663
9	13.97464	13.96449	0.07266211	0.1452714
10	18.96142	18.94989	0.06080146	0.1215659
11	19.55673	19.56059	0.01973859	0.03948109
12	21.96473	21.95933	0.02457163	0.04913721
13	24.24123	24.23828	0.01218646	0.02437143
14	25.94732	25.92175	0.09855004	0.197003
15	28.95591	28.90139	0.1883004	0.3762461
16	31.07021	31.0651	0.01644794	0.03289317
17	33.81524	33.80479	0.03091209	0.06181462
18	35.80227	35.78116	0.05897536	0.1179159
19	38.39277	38.37652	0.04231468	0.08461146
20	39.28811	39.23807	0.1273661	0.2545701



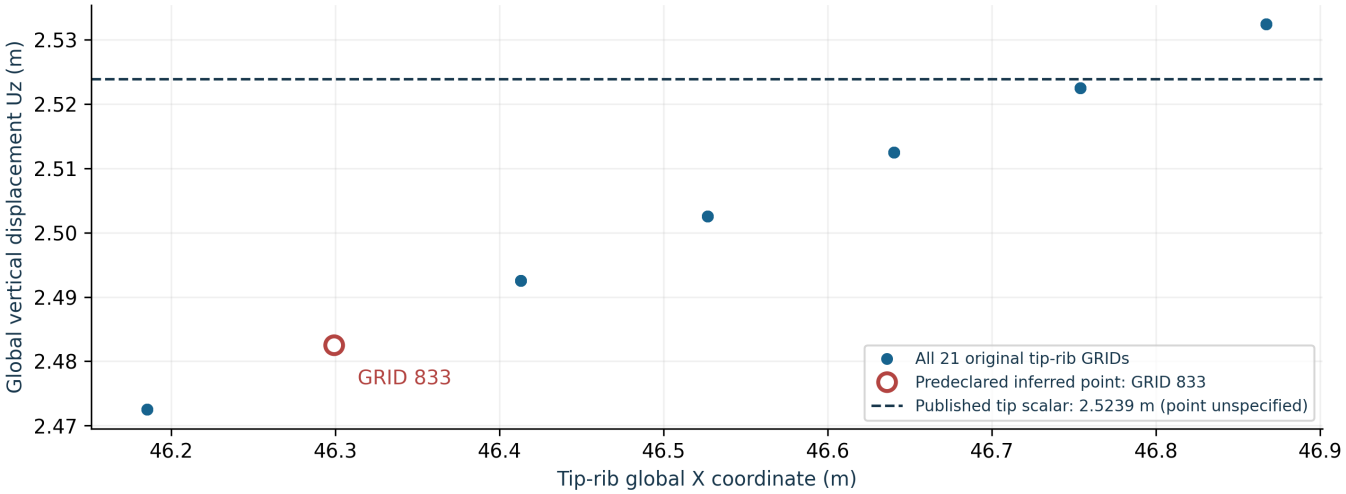
Error = 100|JFEM/reference - 1|. Angular eigenvalues are computed on both sides as (2 pi f)^2; frequency and eigenvalue errors are different measures. The shaded band is the 0.5% comparison criterion.

The workbook provides ordered scalar frequencies, not reference mode vectors. These are ascending-spectrum comparisons; modal identity has not been established by a MAC comparison. Original-pencil residuals and completeness diagnostics accompany the numerical results.

Static tip response and observation point

The published coarse structural tip scalar is 2.5239 m. JFEM gives Uz = 2.4825351 m at predeclared GRID 833. Across all 21 tip-rib nodes, Uz ranges from 2.47251 to 2.53245109 m; the mean is 2.50251708 m.

The authors do not identify a GRID or chord station for the tip scalar. GRID 833 is an inferred maximum-span observation, not an author-specified location. The range is reported without selecting the node that most closely matches the target; this static scalar is not assigned a qualified pass.



Scope and reproducibility

These are independent published computational results, not experimental measurements. The separate coupled aeroelastic displacement of 2.591 m is not the target for this structural-only solve. The medium/fine workbook columns are author reference solutions, not JFEM mesh-convergence runs.

The deck controls and physics are preserved. The reference solver release is unspecified. CQUADR uses the JFEM native TACS-formulation shell operator; retaining a card does not establish formulation identity. No artificial mass, diagonal stiffness or source-constraint removal is used to obtain modes.

Anisotropic, independent-material and MID4 PSHELL force/stress recovery is not currently exported. Static translations and constraint reactions remain available in JSON; absent stresses are not represented as zero. Native modal results can be opened in Boxer.

Reproduce from the shared repository environment:

```
julia --startup-file=no --project=. validation/cases/
CRM_uCRM9_published/reproduce.jl
```

References and provenance: published_structural_references.csv retains all 63 coarse/medium/fine scalar targets and cell addresses; reference_provenance.json binds their workbook and archive-member hashes. curation_receipt.json records the sole software-name comment neutralization; every active card is identical to the source run.

Recorded run: 2026-10-19. Source revision: b74dcdff + reviewed source changes (exact SHA256 in results/run_receipt.json). Full precision: results/comparison.json and metrics.csv. Source hashes: results/run_receipt.json. Stage timings: results/SOL101 and results/SOL103, in the adjacent TIMINGS.JSON files.